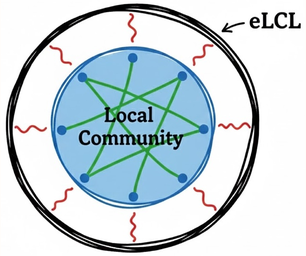

Adaptive Cannistraci-Hebb Network Automata Modelling of Complex Networks for Path-based Link Prediction

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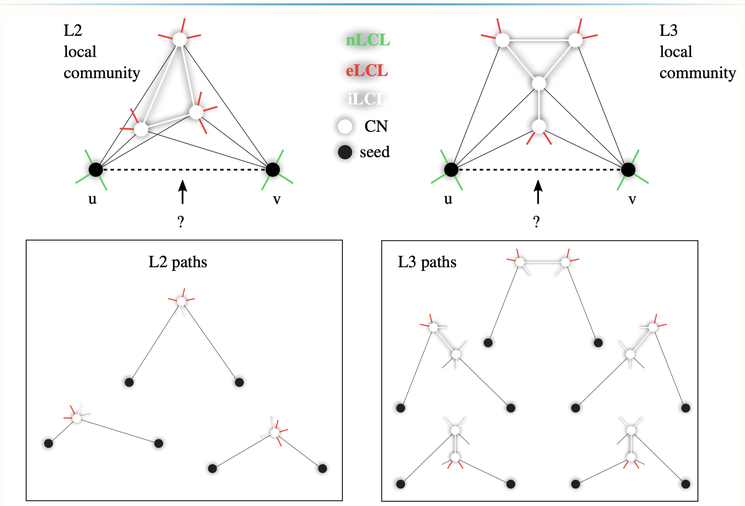
Homepage Paper Code

Cannistraci-Hebb (CH) Theory

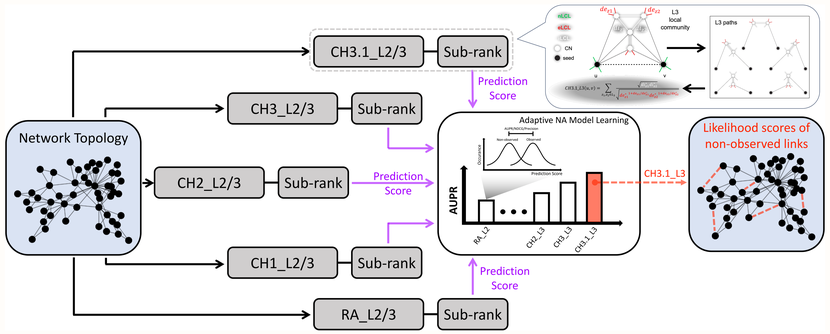


Model	$L2$	$L3$	Ln
RA	$\text{RA-L2}(u,v) = \sum_{z \in L_2} \frac{1}{d_z}$	$\text{RA-L3}(u,v) = \sum_{z_1,z_2 \in L_3} \frac{1}{\sqrt{d_{z_1} d_{z_2}}}$	$\text{RA-Ln}(u,v) = \sum_{z_1,\dots,z_{n-1} \in L_n} \frac{1}{\left(\prod_{k=1}^{n-1} d_{z_k}\right)^{1/(n-1)}}$
CH1	$\text{CH1-L2}(u,v) = \sum_{z \in L_2} \frac{di_z}{d_z}$	$\text{CH1-L3}(u,v) = \sum_{z_1,z_2 \in L_3} \frac{\sqrt{di_{z_1} di_{z_2}}}{\sqrt{d_{z_1} d_{z_2}}}$	$\text{CH1-Ln}(u,v) = \sum_{z_1,\dots,z_{n-1} \in L_n} \frac{\left(\prod_{k=1}^{n-1} di_{z_k}\right)^{1/(n-1)}}{\left(\prod_{k=1}^{n-1} d_{z_k}\right)^{1/(n-1)}}$
CH2	$\text{CH2-L2}(u,v) = \sum_{z \in L_2} \frac{di^*_z}{de^*_z}$	$\text{CH2-L3}(u,v) = \sum_{z_1,z_2 \in L_3} \frac{\sqrt{di^*_{z_1} di^*_{z_2}}}{\sqrt{de^*_{z_1} de^*_{z_2}}}$	$\text{CH2-Ln}(u,v) = \sum_{z_1,\dots,z_{n-1} \in L_n} \frac{\left(\prod_{k=1}^{n-1} di^*_{z_k}\right)^{1/(n-1)}}{\left(\prod_{k=1}^{n-1} de^*_{z_k}\right)^{1/(n-1)}}$
CH3	$\text{CH3-L2}(u,v) = \sum_{z \in L_2} \frac{1}{de^*_z}$	$\text{CH3-L3}(u,v) = \sum_{z_1,z_2 \in L_3} \frac{1}{\sqrt{de^*_{z_1} de^*_{z_2}}}$	$\text{CH3-Ln}(u,v) = \sum_{z_1,\dots,z_{n-1} \in L_n} \frac{1}{\left(\prod_{k=1}^{n-1} de^*_{z_k}\right)^{1/(n-1)}}$
CH3.1	$\text{CH3.1-L2}(u,v) = \sum_{z \in L_2} \frac{di^*_z}{de^{*\,1+de_z/de^*_z}_z}$	$\text{CH3.1-L3}(u,v) = \sum_{z_1,z_2 \in L_3} \frac{\sqrt{di^*_{z_1} di^*_{z_2}}}{\sqrt{de^{*\,1+de_{z_1}/de^*_{z_1}}_{z_1} de^{*\,1+de_{z_2}/de^*_{z_2}}_{z_2}}}$	$\text{CH3.1-Ln}(u,v) = \sum_{z_1,\dots,z_{n-1} \in L_n} \frac{\left(\prod_{k=1}^{n-1} di^*_{z_k}\right)^{1/(n-1)}}{\left(\prod_{k=1}^{n-1} de^{*\,1+de_{z_k}/de^*_{z_k}}_{z_k}\right)^{1/(n-1)}}$

Two pillars:

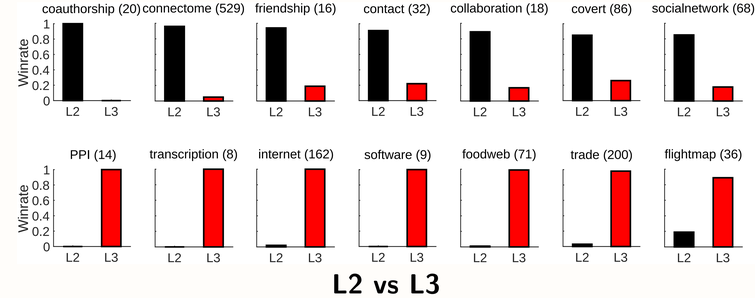
- Minimization of external links within local communities.
- Path-based definition of local communities: homophilic (similarity-driven, L2) vs synergetic (diversity-driven, L3)

Cannistraci-Hebb Network Automata (CHA)

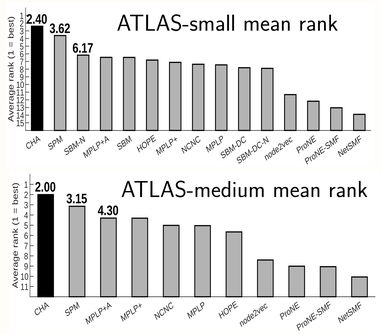


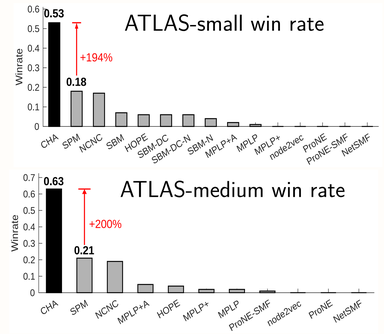
- **Adaptive model selection:** CHA automatically chooses the most suitable Cannistraci–Hebb rule (L2 vs. L3) for each network by evaluating how well candidate models separate observed from non-observed links.
- **Sub-ranking refinement:** When CH scores are tied, CHA refines predictions using a shortest-path–based proximity measure that compares node-pair topological similarity.

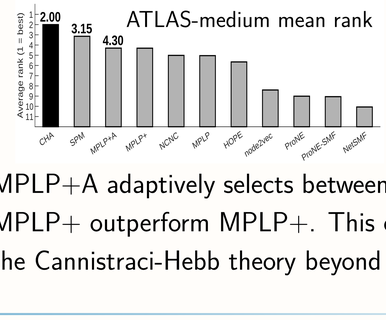
Experiments

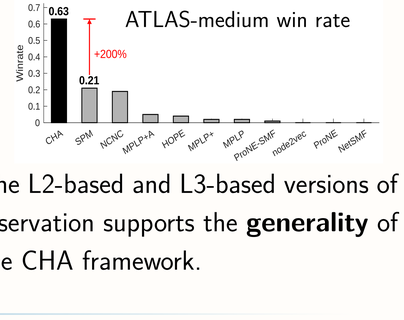


L2 vs L3









MPLP+A adaptively selects between the L2-based and L3-based versions of MPLP+ outperform MPLP+. This observation supports the **generality** of the Cannistraci-Hebb theory beyond the CHA framework.

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